

MICHEL POPOLIN

Senior Software Engineer | Full-Stack & Distributed Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer and Systems Architect with **24+ years of hands-on experience** architecting, modernizing, and scaling high-throughput distributed systems, event-driven microservices, and full-stack applications. Deep expertise in Node.js, TypeScript, GraphQL, and modern web frameworks (React, Vue.js), backed by an enterprise foundation in Java and Ruby on Rails. Proven track record across FinTech, HealthTech, Digital Media at companies like Clearco, PayPal, and Clipboard Health. Holds certifications as a Node.js Developer (JSNSD) and Sun Certified Enterprise Architect (SCEA).

- **Core Competencies:** Distributed Systems Architecture, Full-Stack Engineering, Event-Driven Microservices, API Design (GraphQL & REST) and Technical Leadership.
- **Performance & Systems Optimization:** Specialist in root-cause analysis, V8 runtime internals, resolving ENOMEM/memory leaks, and slashing database response times through N+1 query elimination and execution plan optimization.
- **Quality & Reliability:** Experienced in introducing automated testing standards (Vitest, Playwright, Jest) and continuous monitoring (Grafana, Sentry, Datadog), cutting production bugs by ~40%.

WORK EXPERIENCE

Senior Software Engineer — Clearco

Toronto, ON (Remote) | Apr 2023 – Present

- **Runtime & Frontend Modernization:** Spearheaded enterprise Node.js runtime migration (Node 16 to Node 24) and led frontend architecture modernization from legacy Vue 2 to Vue 3 using Vite, TypeScript (v5.x), Vitest, and Playwright, eliminating EoL security risks and accelerating build speeds.
- **Performance & Latency Optimization:** Slashed critical API response times by ~86% (from ~12s down to ~1.6s) by refactoring legacy pipelines, fixing N+1 PostgreSQL queries, applying targeted indexes, and bounding I/O concurrency.
- **Event-Driven Architecture:** Architected low-latency microservice pipelines utilizing GCP Pub/Sub, RabbitMQ, and Redis for distributed message queuing and asynchronous task orchestration.
- **AI Pipeline Integration:** Engineered LLM-powered backend processing pipelines for automated financial statement analysis, reducing due diligence review times from hours down to under 15 minutes.
- **Quality & Reliability:** Reduced production regressions by ~40% through strict TypeScript typing, comprehensive Vitest/Playwright unit and E2E test suites, and real-time observability via Grafana and Sentry

Technologies: Node.js (v16/v24), TypeScript (v4/v5), Vue.js (v2/v3), Vite, NestJS, GCP Pub/Sub, RabbitMQ, Redis, PostgreSQL, Prisma, Vitest, Playwright, LLM APIs, Docker, Kubernetes, Grafana, Sentry, AWS, GCP.

Senior Software Engineer — Clipboard

San Francisco, CA (Remote) | May 2022 – Apr 2023

- **GraphQL & Microservices:** Architected high-throughput NestJS microservices utilizing GraphQL and MongoDB, designing optimized schemas/resolvers for complex healthcare shift-matching queries.
- **Real-Time Telemetry Pipelines:** Engineered geofencing event pipelines using RabbitMQ and Redis caching to process thousands of concurrent location telemetry events (via Radar.com), automating instant payout calculations.
- **Full-Stack & Mobile Engineering:** Developed cross-platform mobile features (React Native, Ionic, Capacitor) and configured Bitrise CI/CD automation to sync seamlessly with GraphQL backend services.
- **API Modernization:** Led the architectural transition of legacy Express endpoints to modular NestJS microservices in strict TypeScript, implementing company-wide testing standards.

Technologies: Node.js (v16), TypeScript, NestJS, GraphQL, MongoDB, PostgreSQL, RabbitMQ, Redis, Radar.com Telemetry, Vitest, Jest, React Native, Ionic, Capacitor, Bitrise (CI/CD), AWS, Datadog.

Senior Software Engineer — PayPal

San Diego, CA (Remote) | Jan 2022 – Mar 2023

- **Full-Stack & Middleware Layer:** Engineered high-concurrency Node.js/TypeScript BFF middleware services connecting React frontends with core Java payment APIs for global merchant onboarding.
- **High-Concurrency Resilience:** Diagnosed and resolved production memory leaks, race conditions, and async execution bottlenecks across high-traffic microservices under strict security SLAs.

Technologies: Node.js, TypeScript, JavaScript (ES6+), React.js, Java Core APIs, RESTful Microservices, High-Concurrency Systems, BFF Architecture.

Senior Software Engineer — Digital Trends

Portland, OR (Remote) | Feb 2021 – Jul 2022

- **Monolith Refactoring:** Refactored a monolithic media engine into modular Node.js/TypeScript microservices with centralized JWT-based authentication.
- **Search & Ad Platforms:** Integrated GraphQL APIs with Elasticsearch and Redis caching to aggregate multi-catalog product data (Amazon, BestBuy) for high-speed product search and programmatic ad execution.

Technologies: Node.js, TypeScript, GraphQL, React.js, Redis, Elasticsearch, REST APIs, JWT/OAuth, AWS

Tech Lead Engineer — Trix Tecnologia

Brasília, DF, Brazil | Jun 2015 – Feb 2021

- **Architecture Evolution:** Directed backend system architecture for a nationwide clinical suite, leading the evolution of legacy Java and Ruby on Rails monoliths into scalable Node.js microservices.
- **Hardware & Telemetry Integration:** Engineered low-level protocol interfaces and stream processing to interface directly with medical laboratory equipment, diagnostic hardware, and PACS/DICOM imaging servers.
- **Real-Time Systems:** Built WebSocket and event-driven messaging pipelines to synchronize clinical patient queue panels with web and mobile clients in real time.

Technologies: Node.js, TypeScript, Java, Ruby on Rails, Node.js Streams/Buffers, WebSockets, PACS/DICOM Protocols, Laboratory Device Interfaces, GraphQL, REST APIs, SQL.

TECHNICAL SKILLS

Languages & Runtimes: Node.js (v14–v24+, Linux Foundation Certified), TypeScript (v4/v5), JavaScript (ES6+), Java, Ruby on Rails, HTML5/CSS3.

Backend & Architecture: NestJS, Express.js, GraphQL (Apollo Server/Client), RESTful Microservices, Backend-for-Frontend (BFF) Pattern, Event-Driven Architecture, Prisma ORM.

Frontend & Mobile: Vue.js (Vue 2 / Vue 3 & Vite), React, React Native, Next.js, Ionic, Capacitor.

Databases, Caching & Messaging: PostgreSQL (v13–v18), MongoDB, Google Cloud Pub/Sub, RabbitMQ, Redis, Elasticsearch

DevOps & Observability: AWS, GCP, Docker, Kubernetes, Terraform, Grafana, Sentry, Datadog, Auth0, Bitrise (CI/CD)

Testing & Quality: Vitest, Playwright, Jest, Unit & E2E Testing Strategies.

Frontend: React, React Native, Vue.js (Vue 2, Vue 3 & Vite), Next.js, Ionic, Capacitor, Rails.

EDUCATION & CERTIFICATIONS

JSNSD: Node.js Application Services Developer — The Linux Foundation / OpenJS Foundation

SCEA: Sun Certified Enterprise Architect for Java EE — Sun Microsystems

Postgraduate Degree: Distributed Computing and OO Architecture — Universidade de Brasília (UnB), 2005–2007

Bachelor's Degree: Computer Science — Universidade Paulista (UNIP), 2000–2004